

Restriction of meat, fish, and poultry in omnivores improves mood: A pilot randomized controlled trial

Background Omnivorous diets are high in arachidonic acid (AA) compared to vegetarian diets. Research shows that high intakes of AA promote changes in brain that can disturb mood. Omnivores who eat fish regularly increase their intakes of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), fats that oppose the negative effects of AA in vivo. In a recent cross-sectional study, omnivores reported significantly worse mood than vegetarians despite higher intakes of EPA and DHA. This study investigated the impact of restricting meat, fish, and poultry on mood. **Findings** Thirty-nine omnivores were randomly assigned to a control group consuming meat, fish, and poultry daily (OMN); a group consuming fish 3-4 times weekly but avoiding meat and poultry (FISH), or a vegetarian group avoiding meat, fish, and poultry (VEG). At baseline and after two weeks, participants completed a food frequency questionnaire, the Profile of Mood States questionnaire and the Depression Anxiety and Stress Scales. After the diet intervention, VEG participants reduced their EPA, DHA, and AA intakes, while FISH participants increased their EPA and DHA intakes. Mood scores were unchanged for OMN or FISH participants, but several mood scores for VEG participants improved significantly after two weeks. **Conclusions** Restricting meat, fish, and poultry improved some domains of short-term mood state in modern omnivores. To our knowledge, this is the first trial to examine the impact of restricting meat, fish, and poultry on mood state in omnivores.